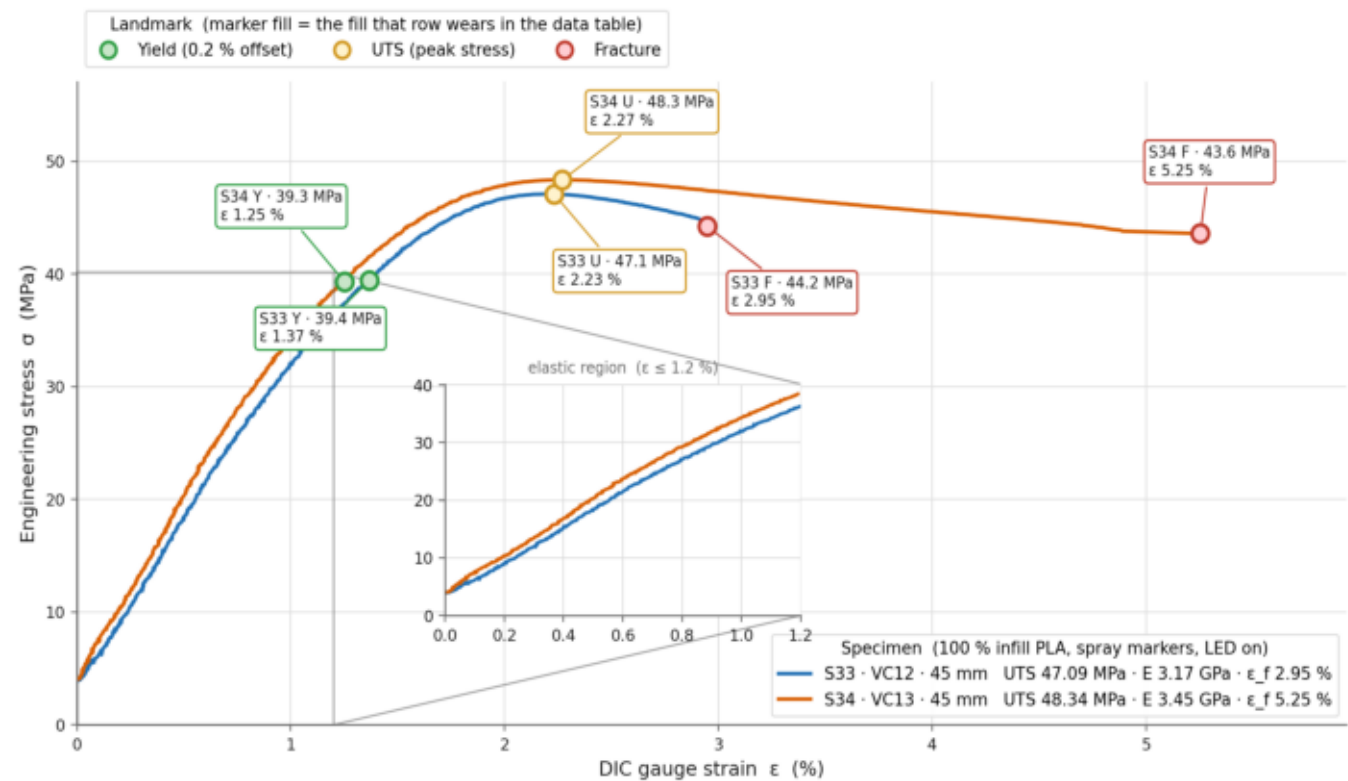


S33 vs S34 — STRESS-STRAIN REFERENCE

Every recorded data point of both frame-capture fracture runs, for line-by-line comparison against extensometer software



HEADLINE NUMBERS

	S33 · VC12 · 45 mm	S34 · VC13 · 45 mm
UTS	47.09 MPa	48.34 MPa
at strain	2.23 %	2.27 %
σ_y (0.2 %)	39.43 MPa	39.31 MPa
at strain	1.37 %	1.25 %
E (fixed window)	3.174 GPa	3.451 GPa
ϵ_f	2.95 %	5.25 %
σ at fracture	44.22 MPa	43.57 MPa
Toughness	1100 kJ/m ³	2184 kJ/m ³
Force anchor	310 N	318 N
Load samples	1134	1185
Captured frames	2371	2200

WHAT IS IN THIS DOCUMENT

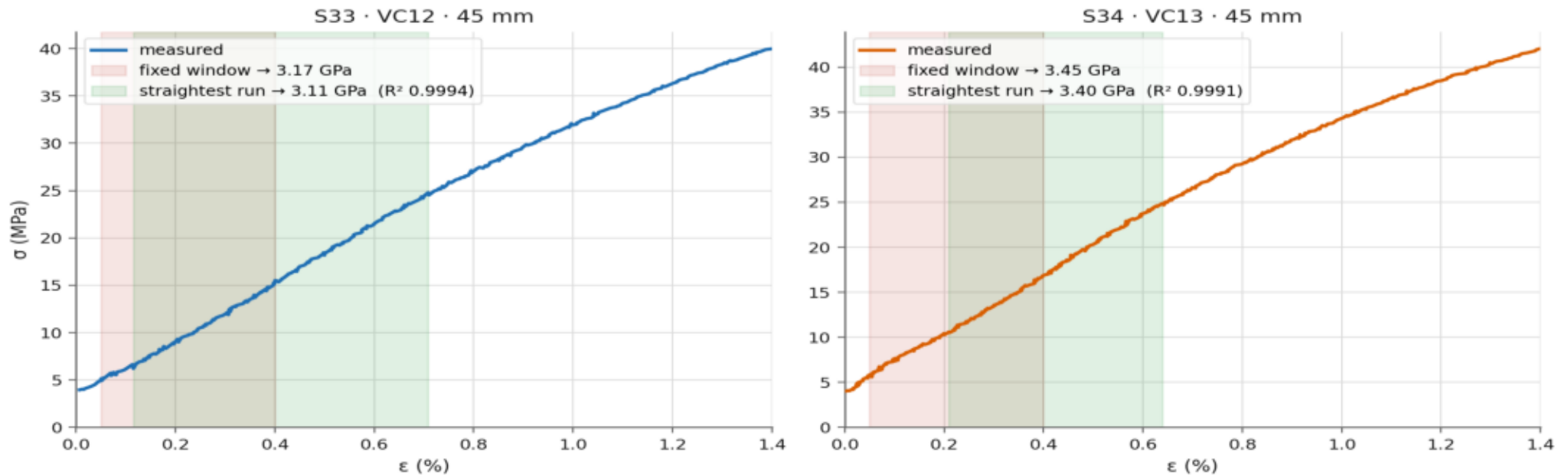
- p2 Why the reported E differs between the two runs — and why σ_y follows it
- p3 COMPARISON TABLE — both runs resampled onto one common strain axis (0.10 % steps), the form to read across
- p4-6 FULL RESOLUTION · S33 — all 570 raw samples in acquisition order
- p7-10 FULL RESOLUTION · S34 — all 824 raw samples in acquisition order

Stress is ENGINEERING stress (force ÷ 80 mm² nominal), anchor-corrected: the preload tared away at the start of the test is added back, which is why these values exceed the tared reading in the CSV header. Strain is DIC gauge strain over the 45 mm / 45 mm marker spacing, zeroed at Px0. Both are exactly what the app's own report button computes — `utm_analysis.analyze()`, no separate maths.

Highlighted rows: ■ Yield (0.2 % offset) ■ UTS (peak stress) ■ Fracture

WHY THE TWO RUNS REPORT A DIFFERENT E

The fit window is fixed; the toe it lands on is not



analyze() fits E over a FIXED 0.05–0.40 % strain window. A specimen that seats more slowly has a longer toe, so that window lands further down the curve and returns a softer modulus. Searching instead for each specimen's own straightest run:

S33 fixed window 3.167 GPa → straightest run 3.113 GPa over 0.12–0.71 % (R^2 0.9994)

S34 fixed window 3.549 GPa → straightest run 3.400 GPa over 0.21–0.64 % (R^2 0.9991)

σ_y is a CONSEQUENCE of this. The 0.2 % offset line has slope E, so a softer E tilts it down, meets the curve later, and pushes σ_y toward UTS: S33 reports σ_y at 86.3 % of its UTS, S34 at 88.5 %. The σ_y spread (39.8 vs 41.7 MPa) is largely the E spread re-expressed.

UTS needs no fit at all and agrees to 2.2 % — it is the number to quote when comparing against the extensometer.

COMPARISON TABLE — COMMON STRAIN AXIS (1/1)

Both runs resampled to 0.10 % strain steps · 60 rows · exact yield / UTS / fracture samples inserted at their true strain

ϵ (%)	S33 σ	S34 σ	Δ	landmark
0.00	—	—	—	
0.10	6.20	5.84	-0.37	
0.20	8.86	8.28	-0.58	
0.30	12.05	10.64	-1.41	
0.40	15.11	13.64	-1.48	
0.50	18.34	17.12	-1.22	
0.60	21.31	20.59	-0.71	
0.70	24.14	24.27	+0.14	
0.80	26.90	27.56	+0.66	
0.90	29.45	30.60	+1.15	
1.00	31.81	33.47	+1.67	
1.10	34.11	35.89	+1.79	
1.20	36.23	38.12	+1.89	
1.25	37.20	37.20	+0.00	S34 yield
1.30	38.25	40.16	+1.92	
1.37	39.51	39.51	+0.00	S33 yield
1.40	40.10	41.78	+1.68	
1.50	41.67	43.13	+1.47	
1.60	43.05	44.27	+1.22	
1.70	44.13	45.18	+1.05	
1.80	44.97	45.88	+0.92	
1.90	45.57	46.39	+0.82	

ϵ (%)	S33 σ	S34 σ	Δ	landmark
2.00	45.94	46.73	+0.79	
2.10	46.11	46.95	+0.84	
2.20	46.16	47.08	+0.92	
2.23	46.14	46.14	+0.00	S33 UTS
2.27	46.12	46.12	+0.00	S34 UTS
2.30	46.07	47.13	+1.06	
2.40	45.94	47.16	+1.22	
2.50	45.75	47.15	+1.39	
2.60	45.54	47.11	+1.57	
2.70	45.31	47.04	+1.73	
2.80	45.10	46.96	+1.86	
2.90	44.87	46.86	+1.99	
2.95	44.77	44.77	+0.00	S33 fracture
3.00	44.66	46.74	+2.08	
3.10	44.45	46.61	+2.17	
3.20	44.23	46.47	+2.24	
3.30	44.05	46.32	+2.28	
3.40	43.85	46.17	+2.32	
3.50	43.67	46.03	+2.36	
3.60	43.47	45.89	+2.42	
3.70	43.27	45.74	+2.47	
3.80	43.06	45.60	+2.54	

ϵ (%)	S33 σ	S34 σ	Δ	landmark
3.90	42.83	45.46	+2.63	
4.00	42.54	45.32	+2.77	
4.10	42.17	45.17	+3.00	
4.20	—	45.03	—	
4.30	—	44.90	—	
4.40	—	44.79	—	
4.50	—	44.66	—	
4.60	—	44.54	—	
4.70	—	44.41	—	
4.80	—	44.29	—	
4.90	—	44.17	—	
5.00	—	44.02	—	
5.10	—	43.89	—	
5.20	—	43.76	—	
5.25	—	—	—	S34 fracture
5.30	—	43.62	—	

σ in MPa. Δ = S34 — S33. “—” means that strain is past that specimen's fracture, so it has no stress there.

Highlighted rows: Yield (0.2 % offset) UTS (peak stress) Fracture

FULL RESOLUTION — S33 · VC12 · 45 mm (1/3)

All 570 recorded samples in acquisition order · UTS 47.09 MPa · σ_y 39.43 MPa · E 3.174 GPa · ϵ_f 2.95 % · 2371 frames captured

#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)
1	0.0074	3.964	54	0.2016	9.168	107	0.3725	14.230	160	0.5444	19.654	213	0.7375	25.475
2	0.0125	4.044	55	0.2025	9.090	108	0.3780	14.487	161	0.5487	19.740	214	0.7391	25.305
3	0.0150	3.995	56	0.2042	9.325	109	0.3806	14.570	162	0.5533	20.000	215	0.7456	25.563
4	0.0185	4.108	57	0.2062	9.011	110	0.3827	14.403	163	0.5561	20.086	216	0.7503	25.911
5	0.0225	4.184	58	0.2084	9.246	111	0.3845	14.745	164	0.5586	20.173	217	0.7546	25.824
6	0.0279	4.278	59	0.2124	9.483	112	0.3861	14.831	165	0.5586	20.259	218	0.7602	26.000
7	0.0324	4.384	60	0.2197	9.635	113	0.3862	14.656	166	0.5590	20.341	219	0.7622	26.086
8	0.0373	4.498	61	0.2255	9.716	114	0.3965	14.999	167	0.5629	20.429	220	0.7654	26.168
9	0.0452	4.846	62	0.2272	9.800	115	0.3980	15.255	168	0.5717	20.519	221	0.7723	26.255
10	0.0497	4.909	63	0.2345	9.877	116	0.3990	15.172	169	0.5751	20.606	222	0.7735	26.342
11	0.0507	5.152	64	0.2365	9.955	117	0.4007	15.517	170	0.5751	20.787	223	0.7817	26.428
12	0.0519	4.963	65	0.2366	10.032	118	0.4071	15.429	171	0.5753	20.869	224	0.7822	26.517
13	0.0524	5.087	66	0.2398	10.112	119	0.4074	15.344	172	0.5802	21.042	225	0.7892	26.604
14	0.0535	5.020	67	0.2428	10.192	120	0.4133	15.607	173	0.5838	20.954	226	0.7914	26.774
15	0.0547	5.215	68	0.2458	10.431	121	0.4140	15.690	174	0.5857	21.294	227	0.7951	27.119
16	0.0588	5.273	69	0.2500	10.516	122	0.4175	15.773	175	0.5915	21.212	228	0.7955	26.859
17	0.0699	5.744	70	0.2558	10.596	123	0.4239	16.033	176	0.5986	21.382	229	0.7996	27.033
18	0.0708	5.473	71	0.2559	10.672	124	0.4264	16.120	177	0.5995	21.472	230	0.8041	27.207
19	0.0756	5.812	72	0.2604	10.837	125	0.4271	16.207	178	0.6012	21.557	231	0.8056	27.293
20	0.0769	5.611	73	0.2613	10.755	126	0.4330	16.290	179	0.6091	21.820	232	0.8119	27.461
21	0.0805	5.543	74	0.2613	10.914	127	0.4343	16.374	180	0.6155	21.909	233	0.8188	27.373
22	0.0806	5.674	75	0.2687	10.993	128	0.4424	16.543	181	0.6172	22.170	234	0.8240	27.637
23	0.0839	5.880	76	0.2726	11.161	129	0.4426	16.632	182	0.6202	22.259	235	0.8267	27.554
24	0.0905	5.945	77	0.2758	11.403	130	0.4440	16.459	183	0.6226	22.348	236	0.8307	27.724
25	0.0974	6.078	78	0.2772	11.321	131	0.4457	16.715	184	0.6292	22.439	237	0.8336	27.980
26	0.1036	6.291	79	0.2818	11.571	132	0.4497	16.978	185	0.6311	22.529	238	0.8389	28.063
27	0.1078	6.433	80	0.2859	11.653	133	0.4555	17.066	186	0.6344	22.614	239	0.8487	28.241
28	0.1093	6.505	81	0.2898	11.734	134	0.4600	17.147	187	0.6427	22.872	240	0.8500	28.328
29	0.1137	6.639	82	0.2949	11.811	135	0.4637	17.315	188	0.6435	22.785	241	0.8507	28.152
30	0.1139	6.573	83	0.3005	11.972	136	0.4666	17.490	189	0.6538	22.961	242	0.8557	28.500
31	0.1152	6.222	84	0.3047	12.144	137	0.4684	17.403	190	0.6550	23.045	243	0.8587	28.414
32	0.1206	6.712	85	0.3063	11.892	138	0.4705	17.579	191	0.6568	23.225	244	0.8593	28.584
33	0.1274	6.923	86	0.3085	12.471	139	0.4723	17.665	192	0.6590	23.133	245	0.8679	28.758
34	0.1344	6.998	87	0.3102	12.225	140	0.4729	17.753	193	0.6608	23.486	246	0.8686	28.840
35	0.1367	7.213	88	0.3109	12.553	141	0.4786	17.840	194	0.6617	23.397	247	0.8738	28.667
36	0.1388	7.073	89	0.3114	12.389	142	0.4868	17.927	195	0.6632	23.309	248	0.8759	29.014
37	0.1414	7.287	90	0.3131	12.637	143	0.4907	18.091	196	0.6694	23.655	249	0.8792	29.095
38	0.1467	7.499	91	0.3201	12.808	144	0.4962	18.267	197	0.6711	23.573	250	0.8830	29.176
39	0.1469	7.356	92	0.3207	12.722	145	0.4978	18.439	198	0.6777	23.740	251	0.8882	29.269
40	0.1474	7.425	93	0.3237	12.891	146	0.4997	18.180	199	0.6886	24.002	252	0.8943	29.355
41	0.1475	7.576	94	0.3291	12.971	147	0.4997	18.355	200	0.6916	24.090	253	0.8965	29.437
42	0.1570	7.798	95	0.3358	13.058	148	0.5047	18.523	201	0.6948	24.267	254	0.8976	29.529
43	0.1603	7.723	96	0.3402	13.312	149	0.5084	18.612	202	0.6950	24.354	255	0.9061	29.869
44	0.1677	8.025	97	0.3437	13.395	150	0.5120	18.787	203	0.6978	24.181	256	0.9086	29.787
45	0.1678	8.251	98	0.3442	13.224	151	0.5139	18.702	204	0.7004	24.442	257	0.9191	29.952
46	0.1682	8.102	99	0.3489	13.477	152	0.5158	19.046	205	0.7043	24.611	258	0.9241	30.133
47	0.1739	8.176	100	0.3510	13.558	153	0.5180	18.958	206	0.7085	24.780	259	0.9243	30.042
48	0.1767	8.552	101	0.3561	13.645	154	0.5191	19.130	207	0.7098	24.523	260	0.9268	30.216
49	0.1775	8.478	102	0.3585	13.977	155	0.5267	19.304	208	0.7207	24.956	261	0.9331	30.297
50	0.1805	8.326	103	0.3588	13.899	156	0.5302	19.219	209	0.7263	25.045	262	0.9336	30.385
51	0.1941	8.856	104	0.3608	13.729	157	0.5359	19.481	210	0.7304	25.221	263	0.9369	30.555
52	0.1956	8.934	105	0.3609	13.812	158	0.5368	19.393	211	0.7322	25.133	264	0.9389	30.470
53	0.1961	8.779	106	0.3723	14.317	159	0.5381	19.569	212	0.7369	25.390	265	0.9514	30.725

Highlighted rows:  Yield (0.2 % offset)

 UTS (peak stress)

 Fracture

Y = yield U = UTS F = fracture

FULL RESOLUTION — S33 · VC12 · 45 mm (2/3)

All 570 recorded samples in acquisition order · UTS 47.09 MPa · σ_y 39.43 MPa · E 3.174 GPa · ϵ_f 2.95 % · 2371 frames captured

#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)
266	0.9515	30.808	319	1.2086	36.447	372	1.4970	41.772	425	1.8303	45.664	478	2.2144	47.085
267	0.9522	30.888	320	1.2090	36.529	373	1.4974	41.632	426	1.8405	45.706	479	2.2153	47.087
268	0.9546	30.978	321	1.2126	36.610	374	1.5024	41.704	427	1.8448	45.750	480	2.2312	47.088 U
269	0.9600	31.148	322	1.2143	36.689	375	1.5041	41.841	428	1.8451	45.793	481	2.2325	47.079
270	0.9622	31.066	323	1.2204	36.768	376	1.5112	41.909	429	1.8512	45.835	482	2.2421	47.074
271	0.9667	31.229	324	1.2292	36.847	377	1.5145	42.052	430	1.8560	45.917	483	2.2505	47.067
272	0.9729	31.315	325	1.2317	37.012	378	1.5184	41.982	431	1.8717	45.994	484	2.2531	47.062
273	0.9899	31.742	326	1.2368	37.169	379	1.5185	42.184	432	1.8734	45.954	485	2.2625	47.056
274	0.9918	31.656	327	1.2435	37.247	380	1.5287	42.315	433	1.8802	46.037	486	2.2773	47.035
275	0.9965	32.084	328	1.2511	37.406	381	1.5315	42.247	434	1.8816	46.074	487	2.2827	47.033
276	0.9978	31.914	329	1.2608	37.483	382	1.5432	42.449	435	1.8857	46.106	488	2.2979	47.024
277	1.0070	32.000	330	1.2636	37.571	383	1.5484	42.514	436	1.8993	46.184	489	2.3115	46.995
278	1.0081	32.170	331	1.2678	37.650	384	1.5584	42.581	437	1.9001	46.145	490	2.3185	46.982
279	1.0124	32.250	332	1.2682	37.728	385	1.5615	42.652	438	1.9033	46.223	491	2.3291	46.968
280	1.0169	32.421	333	1.2718	37.808	386	1.5642	42.718	439	1.9046	46.257	492	2.3304	46.957
281	1.0231	32.505	334	1.2862	38.037	387	1.5767	42.909	440	1.9190	46.293	493	2.3467	46.943
282	1.0270	32.586	335	1.2864	38.117	388	1.5818	42.974	441	1.9283	46.323	494	2.3488	46.930
283	1.0367	32.673	336	1.2949	38.272	389	1.5877	43.103	442	1.9456	46.420	495	2.3577	46.907
284	1.0414	32.839	337	1.3009	38.351	390	1.5978	43.233	443	1.9487	46.454	496	2.3621	46.914
285	1.0416	33.177	338	1.3048	38.505	391	1.5996	43.165	444	1.9539	46.482	497	2.3814	46.861
286	1.0422	33.006	339	1.3094	38.429	392	1.6007	43.302	445	1.9559	46.514	498	2.3857	46.845
287	1.0423	32.925	340	1.3114	38.586	393	1.6059	43.359	446	1.9596	46.544	499	2.3963	46.827
288	1.0497	33.089	341	1.3168	38.740	394	1.6103	43.422	447	1.9647	46.570	500	2.4064	46.807
289	1.0513	33.260	342	1.3188	38.663	395	1.6142	43.483	448	1.9692	46.596	501	2.4099	46.789
290	1.0602	33.422	343	1.3305	38.815	396	1.6217	43.543	449	1.9781	46.625	502	2.4114	46.768
291	1.0729	33.673	344	1.3325	38.886	397	1.6305	43.600	450	1.9825	46.654	503	2.4234	46.753
292	1.0780	33.753	345	1.3503	39.203	398	1.6389	43.664	451	1.9899	46.682	504	2.4262	46.734
293	1.0870	33.836	346	1.3548	39.284	399	1.6535	43.844	452	2.0091	46.752	505	2.4521	46.672
294	1.0891	33.918	347	1.3607	39.362	400	1.6556	43.904	453	2.0154	46.770	506	2.4608	46.657
295	1.0950	33.998	348	1.3618	39.504	401	1.6637	43.962	454	2.0218	46.789	507	2.4698	46.633
296	1.0963	34.081	349	1.3680	39.431 Y	402	1.6694	44.075	455	2.0297	46.811	508	2.4779	46.595
297	1.0983	34.168	350	1.3684	39.584	403	1.6700	44.019	456	2.0352	46.857	509	2.4779	46.610
298	1.1006	34.241	351	1.3763	39.661	404	1.6732	44.130	457	2.0395	46.832	510	2.4885	46.572
299	1.1082	34.323	352	1.3874	39.889	405	1.6802	44.191	458	2.0477	46.874	511	2.4933	46.551
300	1.1171	34.570	353	1.3991	39.964	406	1.6914	44.364	459	2.0667	46.942	512	2.5037	46.530
301	1.1228	34.651	354	1.4019	40.035	407	1.6981	44.421	460	2.0776	46.954	513	2.5265	46.481
302	1.1256	34.738	355	1.4030	40.112	408	1.7029	44.479	461	2.0870	46.964	514	2.5327	46.438
303	1.1284	34.823	356	1.4064	40.184	409	1.7129	44.531	462	2.0919	46.980	515	2.5388	46.413
304	1.1294	34.907	357	1.4079	40.255	410	1.7130	44.583	463	2.0964	46.992	516	2.5521	46.389
305	1.1356	34.986	358	1.4158	40.331	411	1.7219	44.639	464	2.1097	47.007	517	2.5576	46.366
306	1.1393	35.065	359	1.4161	40.409	412	1.7279	44.697	465	2.1145	47.017	518	2.5581	46.342
307	1.1470	35.149	360	1.4230	40.482	413	1.7335	44.751	466	2.1194	47.027	519	2.5621	46.318
308	1.1534	35.233	361	1.4312	40.554	414	1.7459	44.901	467	2.1308	47.045	520	2.5740	46.291
309	1.1612	35.549	362	1.4335	40.778	415	1.7488	44.952	468	2.1327	47.057	521	2.5812	46.269
310	1.1630	35.471	363	1.4353	40.705	416	1.7542	45.004	469	2.1407	47.063	522	2.6006	46.196
311	1.1633	35.630	364	1.4469	40.847	417	1.7644	45.158	470	2.1540	47.066	523	2.6137	46.169
312	1.1695	35.712	365	1.4516	40.920	418	1.7649	45.106	471	2.1559	47.065	524	2.6207	46.146
313	1.1774	35.793	366	1.4529	40.995	419	1.7716	45.206	472	2.1645	47.072	525	2.6314	46.069
314	1.1785	35.881	367	1.4533	41.137	420	1.7837	45.250	473	2.1747	47.079	526	2.6409	46.095
315	1.1804	36.041	368	1.4577	41.068	421	1.7923	45.301	474	2.1752	47.076	527	2.6504	46.044
316	1.1849	35.962	369	1.4667	41.208	422	1.8074	45.486	475	2.1824	47.081	528	2.6542	46.018
317	1.1990	36.284	370	1.4857	41.492	423	1.8107	45.532	476	2.1827	47.084	529	2.6564	45.994
318	1.2031	36.364	371	1.4892	41.563	424	1.8216	45.621	477	2.1982	47.084	530	2.6671	45.965

Highlighted rows: Yield (0.2 % offset)

 UTS (peak stress)

 Fracture

Y = yield U = UTS F = fracture

FULL RESOLUTION — S33 · VC12 · 45 mm (3/3)

All 570 recorded samples in acquisition order · UTS 47.09 MPa · σ_y 39.43 MPa · E 3.174 GPa · ϵ_f 2.95 % · 2371 frames captured

#	ϵ (%)	σ (MPa)
531	2.6817	45.910
532	2.6843	45.883
533	2.6977	45.830
534	2.7011	45.809
535	2.7134	45.780
536	2.7176	45.756
537	2.7264	45.726
538	2.7321	45.698
539	2.7417	45.674
540	2.7461	45.646
541	2.7595	45.615
542	2.7713	45.508
543	2.7744	45.534
544	2.7829	45.478
545	2.7910	45.448
546	2.7916	45.417
547	2.8041	45.386
548	2.8072	45.356
549	2.8222	45.334
550	2.8284	45.306
551	2.8494	45.188
552	2.8569	45.153
553	2.8670	45.120
554	2.8678	45.087
555	2.8732	45.057
556	2.8805	45.028
557	2.8909	45.000
558	2.9053	44.899
559	2.9100	44.867
560	2.9175	44.830
561	2.9196	44.792
562	2.9248	44.761
563	2.9343	44.725
564	2.9388	44.683
565	2.9437	44.644
566	2.9475	44.223 F
567	2.9541	44.560
568	2.9609	44.307
569	2.9611	44.427
570	2.9635	44.473

Highlighted rows: Yield (0.2 % offset) UTS (peak stress) Fracture Y = yield U = UTS F = fracture

FULL RESOLUTION — S34 · VC13 · 45 mm (1/4)

All 824 recorded samples in acquisition order · UTS 48.34 MPa · σ_y 39.31 MPa · E 3.451 GPa · ϵ_f 5.25 % · 2200 frames captured

#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)
1	-0.0024	4.001	54	0.1132	7.987	107	0.2686	12.378	160	0.4089	17.006	213	0.5369	21.736
2	0.0024	4.024	55	0.1186	8.066	108	0.2717	12.549	161	0.4100	17.185	214	0.5411	21.825
3	0.0067	4.092	56	0.1204	8.150	109	0.2738	12.461	162	0.4126	17.360	215	0.5442	21.911
4	0.0092	4.057	57	0.1240	8.227	110	0.2757	12.631	163	0.4139	17.271	216	0.5503	22.002
5	0.0120	4.139	58	0.1240	8.307	111	0.2760	12.808	164	0.4148	17.094	217	0.5542	22.099
6	0.0169	4.271	59	0.1297	8.392	112	0.2787	12.715	165	0.4170	17.450	218	0.5567	22.275
7	0.0170	4.196	60	0.1358	8.556	113	0.2817	12.900	166	0.4222	17.623	219	0.5593	22.190
8	0.0176	4.367	61	0.1372	8.630	114	0.2834	13.068	167	0.4252	17.537	220	0.5643	22.365
9	0.0227	4.483	62	0.1390	8.478	115	0.2838	12.984	168	0.4254	17.802	221	0.5663	22.814
10	0.0241	4.792	63	0.1398	8.716	116	0.2878	13.244	169	0.4287	17.712	222	0.5663	22.904
11	0.0256	4.930	64	0.1447	8.798	117	0.2899	13.151	170	0.4311	17.977	223	0.5670	22.722
12	0.0264	4.597	65	0.1488	8.876	118	0.2957	13.330	171	0.4316	18.064	224	0.5675	22.455
13	0.0264	4.865	66	0.1496	8.958	119	0.2987	13.414	172	0.4329	18.153	225	0.5676	22.540
14	0.0278	5.004	67	0.1541	9.120	120	0.3008	13.585	173	0.4339	17.892	226	0.5694	22.632
15	0.0283	4.702	68	0.1547	9.038	121	0.3012	13.496	174	0.4371	18.244	227	0.5722	22.992
16	0.0317	5.077	69	0.1576	9.284	122	0.3051	13.671	175	0.4382	18.332	228	0.5801	23.168
17	0.0361	5.147	70	0.1617	9.200	123	0.3101	13.757	176	0.4457	18.420	229	0.5822	23.083
18	0.0363	5.226	71	0.1705	9.367	124	0.3129	13.934	177	0.4469	18.601	230	0.5898	23.259
19	0.0374	5.299	72	0.1727	9.533	125	0.3150	13.843	178	0.4499	18.509	231	0.5928	23.350
20	0.0374	5.374	73	0.1731	9.452	126	0.3184	14.023	179	0.4539	18.690	232	0.5946	23.524
21	0.0436	5.521	74	0.1756	9.696	127	0.3208	14.109	180	0.4545	18.780	233	0.5951	23.615
22	0.0460	5.448	75	0.1767	9.776	128	0.3217	14.282	181	0.4551	19.134	234	0.5963	23.436
23	0.0472	5.675	76	0.1797	9.614	129	0.3240	14.192	182	0.4587	18.956	235	0.5973	23.709
24	0.0472	5.748	77	0.1822	9.857	130	0.3274	14.369	183	0.4591	18.867	236	0.6036	23.797
25	0.0509	5.826	78	0.1898	10.022	131	0.3314	14.452	184	0.4598	19.048	237	0.6077	23.889
26	0.0523	5.902	79	0.1907	9.940	132	0.3325	14.538	185	0.4655	19.309	238	0.6104	24.061
27	0.0545	5.602	80	0.1919	10.102	133	0.3333	14.625	186	0.4661	19.221	239	0.6112	24.150
28	0.0556	5.977	81	0.1936	10.183	134	0.3386	14.718	187	0.4701	19.400	240	0.6127	23.975
29	0.0578	6.130	82	0.1995	10.351	135	0.3445	14.894	188	0.4729	19.490	241	0.6192	24.330
30	0.0590	6.053	83	0.1996	10.432	136	0.3451	14.807	189	0.4790	19.760	242	0.6196	24.240
31	0.0609	6.202	84	0.2005	10.268	137	0.3474	14.982	190	0.4834	19.673	243	0.6217	24.419
32	0.0667	6.278	85	0.2099	10.603	138	0.3526	15.069	191	0.4836	19.579	244	0.6279	24.601
33	0.0691	6.584	86	0.2111	10.519	139	0.3544	15.332	192	0.4863	19.844	245	0.6285	24.509
34	0.0695	6.736	87	0.2168	10.685	140	0.3549	15.157	193	0.4873	20.032	246	0.6390	24.865
35	0.0713	6.353	88	0.2197	10.852	141	0.3563	15.243	194	0.4886	19.937	247	0.6407	24.772
36	0.0714	6.503	89	0.2209	10.768	142	0.3564	15.420	195	0.4916	20.123	248	0.6420	24.687
37	0.0747	6.428	90	0.2218	11.018	143	0.3576	15.509	196	0.4941	20.212	249	0.6453	25.051
38	0.0753	6.657	91	0.2222	10.935	144	0.3607	15.599	197	0.5001	20.305	250	0.6486	24.961
39	0.0774	6.816	92	0.2255	11.190	145	0.3634	15.769	198	0.5024	20.387	251	0.6512	25.132
40	0.0786	6.893	93	0.2268	11.356	146	0.3658	15.684	199	0.5058	20.476	252	0.6548	25.302
41	0.0819	7.051	94	0.2285	11.106	147	0.3742	15.860	200	0.5087	20.669	253	0.6564	25.395
42	0.0829	7.126	95	0.2292	11.273	148	0.3809	15.947	201	0.5094	20.752	254	0.6574	25.219
43	0.0830	6.973	96	0.2319	11.436	149	0.3820	16.390	202	0.5095	20.575	255	0.6619	25.486
44	0.0866	7.203	97	0.2413	11.522	150	0.3821	16.029	203	0.5122	20.926	256	0.6673	25.577
45	0.0966	7.281	98	0.2451	11.779	151	0.3826	16.117	204	0.5141	20.840	257	0.6698	25.666
46	0.0969	7.517	99	0.2460	11.609	152	0.3832	16.206	205	0.5152	21.013	258	0.6736	25.752
47	0.0983	7.592	100	0.2469	11.696	153	0.3861	16.302	206	0.5175	21.104	259	0.6801	25.843
48	0.1014	7.358	101	0.2502	11.865	154	0.3889	16.563	207	0.5280	21.287	260	0.6804	26.016
49	0.1019	7.674	102	0.2530	11.948	155	0.3896	16.475	208	0.5284	21.200	261	0.6805	25.932
50	0.1067	7.439	103	0.2546	12.121	156	0.3954	16.651	209	0.5289	21.561	262	0.6840	26.102
51	0.1073	7.755	104	0.2594	12.032	157	0.3975	16.736	210	0.5310	21.470	263	0.6881	26.190
52	0.1091	7.832	105	0.2621	12.208	158	0.3983	16.828	211	0.5329	21.378	264	0.6905	26.367
53	0.1109	7.910	106	0.2633	12.294	159	0.4018	16.916	212	0.5344	21.648	265	0.6932	26.277

Highlighted rows:  Yield (0.2 % offset)

 UTS (peak stress)

 Fracture

Y = yield U = UTS F = fracture

FULL RESOLUTION — S34 · VC13 · 45 mm (2/4)

All 824 recorded samples in acquisition order · UTS 48.34 MPa · σ_y 39.31 MPa · E 3.451 GPa · ϵ_f 5.25 % · 2200 frames captured

#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)
266	0.6975	26.458	319	0.8705	31.029	372	1.0603	35.632	425	1.2718	39.868	478	1.5276	43.874
267	0.6994	26.631	320	0.8777	31.297	373	1.0639	35.711	426	1.2811	40.021	479	1.5289	43.942
268	0.7057	26.546	321	0.8796	31.209	374	1.0677	35.795	427	1.2893	40.096	480	1.5299	44.009
269	0.7065	26.717	322	0.8826	31.380	375	1.0705	35.879	428	1.2927	40.175	481	1.5387	44.143
270	0.7068	26.806	323	0.8859	31.465	376	1.0770	35.962	429	1.2963	40.254	482	1.5423	44.210
271	0.7109	26.895	324	0.8874	31.644	377	1.0806	36.042	430	1.2982	40.335	483	1.5426	44.071
272	0.7141	26.986	325	0.8882	31.557	378	1.0833	36.124	431	1.2983	40.485	484	1.5508	44.272
273	0.7157	27.075	326	0.8958	31.726	379	1.0838	36.210	432	1.3028	40.410	485	1.5594	44.341
274	0.7176	27.163	327	0.8967	31.813	380	1.0922	36.294	433	1.3077	40.566	486	1.5716	44.409
275	0.7238	27.249	328	0.8981	31.902	381	1.0923	36.376	434	1.3118	40.645	487	1.5754	44.471
276	0.7305	27.336	329	0.9042	31.986	382	1.0960	36.459	435	1.3229	40.722	488	1.5837	44.536
277	0.7328	27.429	330	0.9103	32.165	383	1.1032	36.626	436	1.3259	40.801	489	1.5849	44.598
278	0.7354	27.517	331	0.9137	32.074	384	1.1038	36.704	437	1.3330	40.880	490	1.5939	44.727
279	0.7375	27.694	332	0.9161	32.334	385	1.1045	36.543	438	1.3354	40.959	491	1.5955	44.661
280	0.7389	27.606	333	0.9164	32.502	386	1.1124	36.784	439	1.3424	41.038	492	1.5983	44.850
281	0.7413	27.784	334	0.9183	32.249	387	1.1144	36.868	440	1.3440	41.114	493	1.6002	44.790
282	0.7452	27.872	335	0.9254	32.420	388	1.1214	36.956	441	1.3471	41.186	494	1.6095	44.913
283	0.7452	27.960	336	0.9266	32.676	389	1.1310	37.039	442	1.3569	41.264	495	1.6174	45.039
284	0.7470	28.052	337	0.9266	32.589	390	1.1310	37.122	443	1.3571	41.343	496	1.6180	44.978
285	0.7570	28.140	338	0.9338	32.757	391	1.1311	37.360	444	1.3601	41.422	497	1.6284	45.161
286	0.7570	28.226	339	0.9342	32.844	392	1.1314	37.203	445	1.3630	41.499	498	1.6285	45.096
287	0.7659	28.314	340	0.9461	32.935	393	1.1361	37.283	446	1.3722	41.573	499	1.6316	45.227
288	0.7704	28.491	341	0.9489	33.020	394	1.1432	37.443	447	1.3777	41.645	500	1.6363	45.286
289	0.7714	28.402	342	0.9536	33.193	395	1.1454	37.605	448	1.3833	41.719	501	1.6463	45.344
290	0.7716	28.581	343	0.9555	33.106	396	1.1491	37.522	449	1.3892	41.797	502	1.6557	45.406
291	0.7743	28.665	344	0.9563	33.273	397	1.1555	37.693	450	1.3904	41.873	503	1.6575	45.466
292	0.7768	28.754	345	0.9584	33.357	398	1.1628	37.780	451	1.3924	41.945	504	1.6661	45.528
293	0.7784	28.845	346	0.9612	33.521	399	1.1672	37.857	452	1.3974	42.018	505	1.6775	45.584
294	0.7821	28.934	347	0.9671	33.441	400	1.1696	37.938	453	1.4017	42.091	506	1.6821	45.638
295	0.7838	29.022	348	0.9706	33.606	401	1.1746	38.099	454	1.4094	42.163	507	1.6867	45.691
296	0.7854	29.111	349	0.9713	33.690	402	1.1753	38.020	455	1.4185	42.319	508	1.6874	45.752
297	0.7979	29.204	350	0.9824	33.778	403	1.1834	38.180	456	1.4204	42.243	509	1.6914	45.810
298	0.8010	29.292	351	0.9843	33.867	404	1.1843	38.264	457	1.4217	42.390	510	1.6921	45.862
299	0.8068	29.461	352	0.9862	33.954	405	1.1884	38.345	458	1.4288	42.459	511	1.7094	45.915
300	0.8068	29.377	353	0.9888	34.037	406	1.1913	38.428	459	1.4323	42.533	512	1.7116	45.973
301	0.8116	29.548	354	0.9904	34.122	407	1.2022	38.505	460	1.4432	42.607	513	1.7183	46.029
302	0.8156	29.725	355	0.9953	34.211	408	1.2046	38.582	461	1.4435	42.677	514	1.7219	46.080
303	0.8173	29.810	356	0.9990	34.294	409	1.2112	38.659	462	1.4536	42.755	515	1.7264	46.135
304	0.8187	29.638	357	1.0031	34.375	410	1.2117	38.743	463	1.4567	42.827	516	1.7325	46.186
305	0.8278	29.897	358	1.0056	34.463	411	1.2130	38.825	464	1.4579	42.900	517	1.7368	46.234
306	0.8308	29.983	359	1.0101	34.628	412	1.2179	38.903	465	1.4625	42.973	518	1.7451	46.338
307	0.8314	30.073	360	1.0122	34.542	413	1.2208	38.987	466	1.4651	43.040	519	1.7470	46.287
308	0.8365	30.161	361	1.0196	34.712	414	1.2216	39.072	467	1.4708	43.109	520	1.7490	46.392
309	0.8391	30.244	362	1.0256	34.794	415	1.2274	39.151	468	1.4801	43.177	521	1.7662	46.441
310	0.8410	30.503	363	1.0261	35.045	416	1.2357	39.229	469	1.4847	43.250	522	1.7755	46.492
311	0.8411	30.422	364	1.0277	34.874	417	1.2499	39.309 Y	470	1.4861	43.322	523	1.7798	46.542
312	0.8415	30.331	365	1.0302	35.126	418	1.2519	39.463	471	1.4899	43.396	524	1.7837	46.592
313	0.8521	30.676	366	1.0326	34.957	419	1.2521	39.543	472	1.4992	43.535	525	1.7857	46.640
314	0.8523	30.585	367	1.0384	35.211	420	1.2526	39.385	473	1.4992	43.468	526	1.7930	46.683
315	0.8614	30.767	368	1.0464	35.383	421	1.2568	39.621	474	1.5137	43.602	527	1.7960	46.781
316	0.8615	30.946	369	1.0487	35.297	422	1.2684	39.788	475	1.5149	43.671	528	1.7962	46.727
317	0.8640	30.859	370	1.0501	35.465	423	1.2700	39.943	476	1.5212	43.740	529	1.7989	46.831
318	0.8685	31.117	371	1.0538	35.548	424	1.2706	39.705	477	1.5245	43.807	530	1.8116	46.874

Highlighted rows: Yield (0.2 % offset) UTS (peak stress) Fracture Y = yield U = UTS F = fracture

FULL RESOLUTION — S34 · VC13 · 45 mm (3/4)

All 824 recorded samples in acquisition order · UTS 48.34 MPa · σ_y 39.31 MPa · E 3.451 GPa · ϵ_f 5.25 % · 2200 frames captured

#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)	#	ϵ (%)	σ (MPa)
531	1.8182	46.918	584	2.1988	48.317	637	2.6835	47.901	690	3.2222	46.870	743	3.8153	45.811
532	1.8314	47.003	585	2.2091	48.322	638	2.6904	47.883	691	3.2283	46.847	744	3.8405	45.788
533	1.8345	46.962	586	2.2106	48.324	639	2.6971	47.867	692	3.2394	46.831	745	3.8446	45.770
534	1.8358	47.045	587	2.2181	48.323	640	2.7035	47.855	693	3.2525	46.808	746	3.8550	45.754
535	1.8468	47.090	588	2.2325	48.327	641	2.7137	47.836	694	3.2678	46.788	747	3.8651	45.740
536	1.8528	47.133	589	2.2443	48.332	642	2.7265	47.817	695	3.2752	46.771	748	3.8693	45.724
537	1.8631	47.171	590	2.2477	48.333	643	2.7316	47.802	696	3.2952	46.748	749	3.8881	45.705
538	1.8642	47.213	591	2.2560	48.334	644	2.7441	47.787	697	3.2984	46.728	750	3.9063	45.682
539	1.8878	47.256	592	2.2698	48.338 U	645	2.7604	47.766	698	3.3154	46.708	751	3.9133	45.659
540	1.8878	47.295	593	2.2716	48.337	646	2.7684	47.745	699	3.3189	46.687	752	3.9213	45.636
541	1.8967	47.335	594	2.2832	48.337	647	2.7718	47.726	700	3.3342	46.669	753	3.9349	45.620
542	1.8978	47.370	595	2.2864	48.336	648	2.7824	47.710	701	3.3453	46.649	754	3.9438	45.600
543	1.8986	47.446	596	2.2946	48.334	649	2.7942	47.693	702	3.3605	46.621	755	3.9559	45.581
544	1.8996	47.407	597	2.3053	48.332	650	2.8162	47.673	703	3.3711	46.602	756	3.9636	45.563
545	1.9076	47.475	598	2.3101	48.329	651	2.8168	47.655	704	3.3772	46.585	757	3.9782	45.546
546	1.9190	47.510	599	2.3287	48.328	652	2.8360	47.616	705	3.3921	46.565	758	3.9996	45.526
547	1.9300	47.544	600	2.3367	48.326	653	2.8365	47.636	706	3.4006	46.546	759	4.0059	45.509
548	1.9347	47.584	601	2.3414	48.319	654	2.8432	47.597	707	3.4092	46.526	760	4.0213	45.489
549	1.9416	47.619	602	2.3537	48.312	655	2.8542	47.580	708	3.4284	46.508	761	4.0301	45.466
550	1.9478	47.654	603	2.3619	48.305	656	2.8619	47.557	709	3.4319	46.486	762	4.0371	45.448
551	1.9531	47.685	604	2.3711	48.304	657	2.8784	47.535	710	3.4415	46.467	763	4.0542	45.430
552	1.9613	47.713	605	2.3778	48.303	658	2.8862	47.521	711	3.4485	46.446	764	4.0662	45.413
553	1.9682	47.742	606	2.3957	48.282	659	2.8941	47.502	712	3.4651	46.427	765	4.0833	45.392
554	1.9694	47.768	607	2.3957	48.293	660	2.9102	47.484	713	3.4732	46.405	766	4.0894	45.376
555	1.9835	47.798	608	2.4085	48.275	661	2.9252	47.464	714	3.4913	46.386	767	4.1110	45.356
556	1.9884	47.829	609	2.4121	48.266	662	2.9252	47.449	715	3.4984	46.361	768	4.1110	45.334
557	1.9987	47.854	610	2.4171	48.252	663	2.9325	47.426	716	3.5087	46.342	769	4.1237	45.312
558	2.0040	47.911	611	2.4258	48.243	664	2.9479	47.397	717	3.5249	46.323	770	4.1401	45.293
559	2.0050	47.882	612	2.4345	48.237	665	2.9483	47.378	718	3.5382	46.304	771	4.1566	45.273
560	2.0124	47.937	613	2.4518	48.229	666	2.9638	47.356	719	3.5512	46.287	772	4.1695	45.257
561	2.0176	47.959	614	2.4625	48.219	667	2.9816	47.338	720	3.5539	46.269	773	4.1776	45.242
562	2.0267	47.982	615	2.4728	48.195	668	2.9932	47.319	721	3.5639	46.250	774	4.1831	45.223
563	2.0358	48.004	616	2.4736	48.206	669	2.9997	47.304	722	3.5761	46.227	775	4.2033	45.197
564	2.0432	48.026	617	2.4850	48.184	670	3.0131	47.278	723	3.5977	46.206	776	4.2068	45.178
565	2.0439	48.051	618	2.4902	48.177	671	3.0175	47.260	724	3.6084	46.185	777	4.2198	45.164
566	2.0554	48.074	619	2.5026	48.168	672	3.0377	47.244	725	3.6103	46.166	778	4.2392	45.141
567	2.0669	48.093	620	2.5162	48.153	673	3.0393	47.218	726	3.6282	46.146	779	4.2564	45.119
568	2.0730	48.108	621	2.5238	48.140	674	3.0519	47.196	727	3.6346	46.127	780	4.2676	45.098
569	2.0792	48.129	622	2.5413	48.123	675	3.0679	47.174	728	3.6524	46.108	781	4.2725	45.080
570	2.0832	48.151	623	2.5462	48.110	676	3.0816	47.158	729	3.6639	46.087	782	4.2829	45.060
571	2.0867	48.166	624	2.5499	48.098	677	3.0834	47.140	730	3.6669	46.066	783	4.2945	45.042
572	2.1038	48.180	625	2.5586	48.089	678	3.0917	47.116	731	3.6784	46.048	784	4.3114	45.020
573	2.1104	48.195	626	2.5720	48.076	679	3.1003	47.095	732	3.6928	46.025	785	4.3292	44.993
574	2.1208	48.209	627	2.5810	48.056	680	3.1135	47.076	733	3.7082	46.006	786	4.3399	44.975
575	2.1272	48.223	628	2.5963	48.045	681	3.1284	47.056	734	3.7179	45.989	787	4.3480	44.956
576	2.1331	48.239	629	2.6054	48.035	682	3.1431	47.038	735	3.7331	45.949	788	4.3602	44.931
577	2.1383	48.251	630	2.6109	48.016	683	3.1466	47.018	736	3.7378	45.968	789	4.3777	44.911
578	2.1485	48.261	631	2.6135	48.002	684	3.1589	46.993	737	3.7513	45.931	790	4.3973	44.888
579	2.1571	48.271	632	2.6197	47.989	685	3.1607	46.966	738	3.7685	45.911	791	4.4103	44.865
580	2.1631	48.280	633	2.6303	47.974	686	3.1731	46.947	739	3.7745	45.893	792	4.4136	44.846
581	2.1673	48.288	634	2.6449	47.954	687	3.1899	46.930	740	3.7811	45.876	793	4.4248	44.824
582	2.1822	48.297	635	2.6626	47.937	688	3.2058	46.913	741	3.7924	45.853	794	4.4377	44.804
583	2.1916	48.307	636	2.6764	47.916	689	3.2159	46.891	742	3.8070	45.833	795	4.4602	44.781

Highlighted rows:  Yield (0.2 % offset)

 UTS (peak stress)

 Fracture

Y = yield U = UTS F = fracture

FULL RESOLUTION — S34 · VC13 · 45 mm (4/4)

All 824 recorded samples in acquisition order · UTS 48.34 MPa · σ_y 39.31 MPa · E 3.451 GPa · ϵ_f 5.25 % · 2200 frames captured

#	ϵ (%)	σ (MPa)
796	4.4720	44.755
797	4.4781	44.734
798	4.4923	44.710
799	4.5041	44.691
800	4.5274	44.665
801	4.5385	44.639
802	4.5545	44.617
803	4.5618	44.591
804	4.5843	44.562
805	4.5936	44.534
806	4.6094	44.509
807	4.6247	44.482
808	4.6397	44.449
809	4.6592	44.421
810	4.6737	44.395
811	4.6964	44.361
812	4.7022	44.328
813	4.7045	44.300
814	4.7246	44.268
815	4.7441	44.227
816	4.7600	44.189
817	4.7687	44.147
818	4.7975	44.105
819	4.8072	44.060
820	4.8234	44.007
821	4.8455	43.945
822	4.8808	43.875
823	4.8898	43.795
824	5.2536	43.574 F

Highlighted rows: Yield (0.2 % offset) UTS (peak stress) Fracture Y = yield U = UTS F = fracture